

FIONA CALLAHAN

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PROFILE

- Computational Biology Ph.D. student at University of California, Berkeley working in Dr. Rasmus Nielsen's lab
- Currently working on computational methods development for ecological modeling using data from ancient environmental DNA
- Interested in pedagogy, mentorship, and equity-oriented teaching practices

EDUCATION

PhD	University of California, Berkeley (in progress) Computational Biology	Expected May 2027
BA	Scripps College, GPA 3.9 Mathematical/Computational Biology Major through Harvey Mudd College	May 2021

PUBLICATIONS

Callahan, F. M., Kjær, K. H., Ramsøe, A. D., Willerslev, E., & Nielsen, R. Reproducibility and normalization in ancient environmental DNA. (in preparation).

Kjær, K. H.[†], Ruter, A. H.[†], Menendez-Serra, M.[†], Vogel, N. A.[†], Ramsøe, A. D.[†], Farnsworth, W. R.[†], Siggaard-Andersen, M.-L.[†], Huang, Z.[†], **Callahan, F.**[†], Mesny, F.[†], ... Willerslev, E. (in review). Environmental DNA reveals Reykjavík's human and ecological history.

Callahan, F. M.[†], & Evensen, C.[†] (2026). Co-occurrence networks can preserve emergent properties of ecological communities. *bioRxiv*, 2026-04. DOI:10.64898/2026.04.15.718781.

Li, J. K., Lim, W., **Callahan, F. M.**, Raskin, L. Y., Lemmon-Kishi, M., & Nielsen, R. (2026). STEM-LM: Spatio-Temporal Ecological Modeling via Masked Language Model for Joint Species Distribution. *bioRxiv*, 2026-05.

Callahan, F. M., Li, J. K., & Nielsen, R. (2025). Challenges in detecting ecological interactions using sedimentary ancient DNA data. *Environmental DNA*, 7(2), e70067. DOI:10.1002/edn3.70067.

[†]Joint first authors.

RESEARCH EXPERIENCE

Network modeling for community ecology	Spring 2025 - Present
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UC Berkeley

Advisor: Dr. Rasmus Nielsen

- Designed simulations to benchmark species interaction network modeling tools.
- Explored how co-occurrence networks and species interaction networks are related through generalized Lotka-Volterra models.

Ecological community modeling using ancient eDNA

Spring 2022 - Present

UC Berkeley

Advisor: Dr. Rasmus Nielsen

- Developed mechanistic simulation framework for ecological data from ancient environmental DNA (aeDNA)
- Analysis of aeDNA data from Iceland (in progress)
- Development of new data analysis tools for aeDNA (in progress)
- Development of rigorous statistical frameworks for normalization and use of replicates for aeDNA data

Time series modeling of phytoplankton blooms in the San Francisco Bay

Spring 2022

UC Berkeley (first year PhD rotation project)

Advisor: Dr. Perry de Valpine

- Rotation project (10 weeks) analyzing time series data to predict phytoplankton blooms in the San Francisco Bay

Disease ecology in bees

Fall 2021

UC Berkeley (first year PhD rotation project)

Advisor: Dr. Mike Boots

- Rotation project (10 weeks) building a simulation of disease transmission in bees between different species

Connecting molecular differences in TTN gene to muscle performance

2020 - 2021

WM Keck Science Department, Claremont Colleges

Advisor: Dr. Findley Finseth

- Undergraduate senior thesis research
- Used a computational tool to characterize genetic differences in the protein Titin and connect them to within-species differences in muscle performance

Development of computational annotation tool for the protein titin

Summer 2020

WM Keck Science Department, Claremont Colleges

Funding from Johnson Summer Research Fellowship

Advisor: Dr. Findley Finseth

- Adapted annotation tool for a region of the giant sarcomeric protein titin
- Extensive computational work in Python, R, and Bash

Differential gene expression in pair-bonded versus naïve zebra finch

Spring 2020

WM Keck Science Department, Claremont Colleges

Advisor: Dr. Findley Finseth

- Analyzed RNA-seq data from zebra finch brain tissue

- Performed a comparative analysis of computational tools for RNA-seq differential expression analysis

Differential gene expression in high-running versus control mice Summer 2019

WM Keck Science Department, Claremont Colleges

Advisors: Dr. Findley Finseth, Dr. Jenna Monroy

- Performed RNA extraction from mouse muscle tissue samples
- Developed RNA-seq data analysis pipeline using comparisons of tools for alignment and differential expression analysis
- Worked extensively with Bash shell scripting and various R packages

TEACHING AND MENTORSHIP

Math 50B study group leader, Mount Tamalpais College Summer 2025

- Volunteered as a weekly instructor for a college preparatory math class
- Designed lessons, assignments, and exams

Graduate student instructor, UC Berkeley Summer 2025

- Led three discussion sections of 25-30 students each for *IB35AC: Human Biological Variation*
- Designed lessons, graded student work, and coordinated accommodations for students with disabilities
- Supervised by Dr. Rasmus Nielsen

Research mentor, UC Berkeley Spring 2025 - Summer 2026

- Mentored five undergraduates in research projects
- Designed projects, advertised positions, and selected students
- Met with students weekly to discuss research progress and other mentorship topics

Substitute math teaching and tutoring, Mount Tamalpais College Fall 2024 – Present

- Volunteered as a substitute math teacher and tutor at Mount Tamalpais College

Math 50B instructor, Mount Tamalpais College Summer 2024

- Instructed a pre-college math course at Mount Tamalpais College to help adult students prepare for college-level math courses
- Volunteer position with Mount Tamalpais College, which operates inside of San Quentin Rehabilitation Center

Certificate in Teaching and Learning in Higher Education Spring 2023 - Present

- Certificate program includes attending workshops, pedagogy classes, and working with professors on teaching skills

Graduate student instructor, UC Berkeley Fall 2023, Spring 2024

- Taught *Berkeley Connect in Computational Biology*
- Two sections of 20 undergraduate students each

- Wrote original syllabus, designed class activities, conducted weekly lectures
- Conducted individual mentorship meetings with students

Mentorship Committee organizer, UC Berkeley Fall 2023 - Present

- Organized mentorship and onboarding activities for first-year PhD students
- Organized 2-3 workshops for early-career PhD students each semester
- Paired students with mentors in the program and facilitated mentorship meetings

Python Bootcamp teaching assistant, UC Berkeley Spring 2024

- Assisted in teaching one-week Python coding bootcamp for beginners

Expanding Your Horizons volunteer, UC Berkeley Spring 2023, 2024, 2025, 2026

- 2023: Volunteer workshop assistant for conference to introduce 5th-8th grade girls* to careers in STEM (*Including cis girls, trans youth, gender non-conforming and/or non-binary youth)
- 2024: Workshop leader for conference
- 2025: Workshop leader for conference
- 2026: Workshop leader for conference

Research mentor, UC Berkeley Spring 2022 – Summer 2023

- Mentoring an undergraduate research assistant
- Overseeing mentee's research project and meeting weekly to discuss progress
- Assisting mentee with graduate school applications

Be A Scientist volunteer mentor, King Middle School Spring 2023

- Mentoring 7th grade students at a local middle school for 6-week program
- Students design and conduct their own experiments under mentorship of graduate student volunteers

Mathematics tutor, Scripps College Fall 2017 – Spring 2021

- Individual tutoring and drop-in tutoring center for college students
- Remote tutoring during COVID-19 pandemic
- Single- and multi-variable calculus, Linear algebra, Discrete mathematics

Math team founder and coach, Langley Middle School Fall 2015 - Spring 2017

- Coached two math teams of high-performing middle school students as a volunteer
- Established learning objectives and designed activities

COURSEWORK AND SKILLS

- **Teaching workshops and courses:**
 - Ally in bCourses: Your Key to Accessible Course Content Workshop, Syllabus and Course Design Workshop, Developing a statement of teaching philosophy and teaching portfolio workshop, Assessing Teaching and Learning, How Students Learn, Creating Inclusive Classrooms

- GSPDP301: Mentoring in Higher Education (Spring 2025)
- GSPDP375: Mentoring in Higher Education (Spring 2026)
- **Computer Science:** Data Structures and Program Development, Algorithms, Introduction to Computer Science, Principles of Computer Science
 - **Programming languages:** Python (advanced), R (advanced), Bash (advanced), Java, C++
 - **Lab server administrator:** managed technical support for two Ubuntu servers for the Nielsen lab
- **Statistics:** Introduction to Statistical Computing (graduate level), Hierarchical Statistical Modeling in Environmental Science (graduate level), Statistical models: Theory and Applications (graduate level), Stochastic Processes, Introduction to Statistics at an Advance Level (graduate level), Introduction to Probability at an Advance Level (graduate level)
- **Ecology and Biology:** Network approaches to infer the resilience of ecological communities, Genomics and Bioinformatics, Biostatistics, Molecular Genetics, Ecology and Environmental Biology, Introduction to Mathematical/Computational Biology, Advanced Computational Biology, Evolutionary Genomics, Topics in Biochemistry and Molecular Biology, Molecular/Cellular Biology Lab, Experimental Biology Lab
 - **Lab skills:** RNA extractions, PCR, qPCR, primer design, gel electrophoresis, bacterial transformations
- **Mathematics:** Number Theory, Real Analysis, Linear Algebra, Discrete Mathematics, Multivariable Calculus

PRESENTATIONS AND POSTERS

Exploring how co-occurrence networks preserve emergent properties of ecological communities

Ecological Society of America annual meeting, Salt Lake City, UT

July 2026

Methods for understanding community ecology using sedimentary ancient DNA data

Center for Ancient Evolutionary Genomics symposium, Fredensborg, Denmark

April 2025

Demystifying the path after undergrad

Oral presentation, CSU Fullerton

March 2025

The power of computational biology: Exploring the world we live in

Oral presentation, CCAM seminar, CSU Fullerton

March 2025

Challenges in detecting ecological interactions using sedimentary ancient DNA data

Oral presentation, Section for GeoGenetics, University of Copenhagen

March 2025

Poster presentation, Bay Area Population Genetics

November 2024

Oral presentation, Center for Computational Biology retreat, UC Berkeley

October 2024

Oral presentation, Center for Computational Biology student seminar

September 2024

Oral presentation, Center for Computational Biology orientation August 2024

Simulation study: Which features of ancient environmental DNA data cause incorrect inferences in species distribution models?

Poster presentation, sedaDNA meeting, Potsdam, Germany June 2023

Features of ancient environmental DNA data cause incorrect inferences in species distribution models

Poster presentation, Center for Computational Biology, UC Berkeley November 2023

Oral presentation, Globe Institute Symposium, University of Copenhagen June 2023

Methods comparison for ecological community modeling from ancient environmental DNA data

Poster presentation, Center for Computational Biology retreat, UC Berkeley October 2022

Connecting Molecular Differences in the Protein Titin to Differences in Muscle Performance

Video presentation, Johnson Research Fellowship, Scripps College September 2020

Differential Gene Expression in Muscle Tissue of High-running versus Control Mice

Poster presentation, Keck Science Department, Claremont Colleges September 2019

Investigating the Genetic Basis of Voluntary Exercise in Mice

Oral presentation, Keck Science Department, Claremont Colleges June 2019

HONORS AND AWARDS

Best Senior Thesis Award, Harvey Mudd College Biology Dept. Spring 2021

Johnson Summer Research Fellowship, Scripps College Summer 2020

James E. Scripps Scholar, Scripps College Fall 2017 – Spring 2021
Four-year, half-tuition merit scholarship

Dean's List, Scripps College Fall 2017, Spring 2018, Fall 2018, Spring 2019, Fall 2019

OTHER

Wilderness First Aid (WFA) certified

Spanish: Intermediate Listener, Intermediate Speaker, Advanced Reading and Writing